

# 《线性代数与解析几何》(A)试卷(16-17年度第1学期)

一、(15分) 填空题.

1.  $-\frac{5}{13}$       2.  $(-1)^{mn}ab$       3.  $\begin{pmatrix} 0 & 0 & 0 \\ 2^{n-2} & 0 & 2^{n-2} \\ 0 & 0 & 0 \end{pmatrix}$ , 此题难, 大多数  
元素正确时酌情给分

4. 40      5.  $-1 < k < 0$  .只算出  $-1 < k$  或  $k < 0$  酌情给分

二、(15分) 选择题:

1. D      2. A      3. B      4. B      5. C

三、(8分)计算行列式:

$$D = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 + 1 & x_2 + 1 & x_3 + 1 & \cdots & x_n + 1 \\ x_1^2 + x_1 & x_2^2 + x_2 & x_3^2 + x_3 & \cdots & x_n^2 + x_n \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_1^{n-1} + x_1^{n-2} & x_2^{n-1} + x_2^{n-2} & x_3^{n-1} + x_3^{n-2} & \cdots & x_n^{n-1} + x_n^{n-2} \end{vmatrix}.$$

解

$$D = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ x_1 & x_2 & x_3 & \cdots & x_{n-1} & x_n \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_{n-1}^2 & x_n^2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ x_1^{n-2} & x_2^{n-2} & x_3^{n-2} & \cdots & x_{n-1}^{n-2} & x_n^{n-2} \\ x_1^{n-1} & x_2^{n-1} & x_3^{n-1} & \cdots & x_{n-1}^{n-1} & x_n^{n-1} \end{vmatrix} \quad (+4\text{分})$$

$$= \prod_{1 \leq j < i \leq n} (x_i - x_j) \quad (+4\text{分})$$

四、(15分)

解

$$D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = \begin{vmatrix} \lambda+2 & 1 & 1 \\ \lambda+2 & \lambda & 1 \\ \lambda+2 & 1 & \lambda \end{vmatrix} = (\lambda+2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} \quad (+5\text{分})$$

$$= (\lambda+2) \begin{vmatrix} 1 & 1 & 1 \\ 0 & \lambda-1 & 0 \\ 0 & 0 & \lambda-1 \end{vmatrix} = (\lambda-1)^2(\lambda+2)$$

(1) 当  $D \neq 0$ , 即  $\lambda \neq 1$  且  $\lambda \neq -2$  时, 方程组有唯一解。

$$D_1 = \begin{vmatrix} -2 & 1 & 1 \\ -2 & \lambda & 1 \\ -2 & 1 & \lambda \end{vmatrix} = \begin{vmatrix} -2 & 1 & 1 \\ 0 & \lambda-1 & 0 \\ 0 & 0 & \lambda-1 \end{vmatrix} = -2(\lambda-1)^2, \quad D_2 = \begin{vmatrix} \lambda & -2 & 1 \\ 1 & -2 & 1 \\ 1 & -2 & \lambda \end{vmatrix} = \begin{vmatrix} \lambda-1 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & \lambda-1 \end{vmatrix} = -2(\lambda-1)^2$$

$$D_3 = \begin{vmatrix} \lambda & 1 & -2 \\ 1 & \lambda & -2 \\ 1 & 1 & -2 \end{vmatrix} = \begin{vmatrix} \lambda-1 & 0 & 0 \\ 0 & \lambda-1 & 0 \\ 1 & 1 & -2 \end{vmatrix} = -2(\lambda-1)^2, \quad \text{得} \begin{cases} x_1 = \frac{D_1}{D} = \frac{-2}{\lambda+2} \\ x_2 = \frac{D_2}{D} = \frac{-2}{\lambda+2} \\ x_3 = \frac{D_3}{D} = \frac{-2}{\lambda+2} \end{cases}$$

此时的唯一解是:  $(x_1, x_2, x_3) = \left( \frac{-2}{\lambda+2}, \frac{-2}{\lambda+2}, \frac{-2}{\lambda+2} \right)$  (+3分)

(2) 当  $\lambda = -2$  时,

$$\tilde{A} = \left( \begin{array}{ccc|c} -2 & 1 & 1 & -2 \\ 1 & -2 & 1 & -2 \\ 1 & 1 & -2 & -2 \end{array} \right) \rightarrow \left( \begin{array}{ccc|c} -2 & 1 & 1 & -2 \\ 1 & -2 & 1 & -2 \\ 1 & 1 & -2 & -2 \end{array} \right) \rightarrow \left( \begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -6 \end{array} \right)$$

$r(A) < r(\tilde{A})$ , 此时方程组无解 (+3分)

(3) 当  $\lambda = 1$  时,

$$\tilde{A} = \left( \begin{array}{ccc|c} 1 & 1 & 1 & -2 \\ 1 & 1 & 1 & -2 \\ 1 & 1 & 1 & -2 \end{array} \right) \rightarrow \left( \begin{array}{ccc|c} 1 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$r(A) = r(\tilde{A}) = 1 < 3$ , 此时方程组有无穷多个解。

$$\text{由 } x_1 + x_2 + x_3 = -2, \quad \text{得} \begin{cases} x_1 = -k_1 - k_2 - 2 \\ x_2 = k_1 \\ x_3 = k_2 \end{cases}$$

$$\text{通解为 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = k_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}, \quad k_1, k_2 \text{ 是任意常数} \quad (+4\text{分})$$

五、 (15 分)

解:

$$(\varepsilon_1^T \quad \varepsilon_2^T \quad \varepsilon_3^T \mid \eta_1^T \quad \eta_2^T \quad \eta_3^T) = \left( \begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 2 & 0 \\ 1 & 1 & -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 0 & 3 & -1 \end{array} \right) \rightarrow \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & 2 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} \end{array} \right)$$

$$\text{过渡矩阵 } C = \begin{pmatrix} \frac{1}{2} & 2 & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \quad (8\text{分})$$

$$\text{令 } \xi = x_1 \eta_1 + x_2 \eta_2 + x_3 \eta_3$$

$$\tilde{A} = (\eta_1^T, \eta_2^T, \eta_3^T \mid \xi^T) = \left( \begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 1 & 1 & 1 & 5 \\ 0 & 3 & -1 & 0 \end{array} \right) \xrightarrow{\text{行初等变换}} \left( \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array} \right) \quad (7\text{分})$$

$$\text{得 } x_1 = 1, \quad x_2 = 1, \quad x_3 = 3$$

$\xi$  在  $\eta_1, \eta_2, \eta_3$  下的坐标是  $(1, 1, 3)$

此题有不同的计算方法

六、 (10分)

解:

$$\alpha \times \beta = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \vec{i} + \vec{j} - 3\vec{k} \quad (+7\text{分})$$

$$\text{平面的点法式方程: } (x-1) + (y-0) - 3(z+1) = 0$$

$$\text{即: } x + y - 3z - 4 = 0 \quad (+3\text{分})$$

七、(15 分)  
解

$$|\lambda E - A| = \begin{vmatrix} \lambda-3 & 2 & 4 \\ 2 & \lambda-6 & 2 \\ 4 & 2 & \lambda-3 \end{vmatrix} = (\lambda-7)^2(\lambda+2) \stackrel{\text{令}}{=} 0 \quad (+5\text{分})$$

$A$ 的特征值:  $\lambda = -2, \lambda = 7$  (2重)

对  $\lambda = -2$ :

$$-2E - A = \begin{pmatrix} -5 & 2 & 4 \\ 2 & -8 & 2 \\ 4 & 2 & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} x_1 - 2x_2 = 0 \\ 2x_2 - x_3 = 0 \end{cases} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = k \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \text{得对应 } \lambda = -2 \text{ 的特征向量 } \alpha_1 = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

对  $\lambda = 7$ :

(+5分)

$$7E - A = \begin{pmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$2x_1 + x_2 + 2x_3 = 0 \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = k_1 \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + k_2 \begin{pmatrix} 0 \\ 6 \\ -3 \end{pmatrix}, \text{得对应 } \lambda = 7 \text{ 的特征向量 } \alpha_2 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 0 \\ 6 \\ -3 \end{pmatrix}$$

对特征向量  $\alpha_2 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 0 \\ 6 \\ -3 \end{pmatrix}$  作施密特正交化:

(+3分)

$$\eta_2 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \quad \eta_3 = \begin{pmatrix} 0 \\ 6 \\ -3 \end{pmatrix} - \frac{18}{9} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$$

$$\text{对 } \alpha_1, \eta_2, \eta_3 \text{ 单位化: } \gamma_1 = \frac{1}{\|\alpha_1\|} \alpha_1 = \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}, \quad \gamma_2 = \frac{1}{\|\eta_2\|} \eta_2 = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{-2}{3} \end{pmatrix}, \quad \gamma_3 = \frac{1}{\|\eta_3\|} \eta_3 = \begin{pmatrix} \frac{-2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$$

$$\text{得到正交矩阵 } T = (\gamma_1 \quad \gamma_2 \quad \gamma_3) = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{-2}{3} & \frac{1}{3} \end{pmatrix},$$

正交矩阵  $T$  的答案有无穷多种

$$\text{使得 } T^{-1}AT = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

(+2分)

八、(7 分)

证：设  $AX = 0$  的基础解系为  $\eta_1, \eta_2, \dots, \eta_{n-r}$ ，其中  $r = r(A)$ 。

又设线性无关向量组  $\alpha_1, \alpha_2, \dots, \alpha_s$  与  $\eta_1, \eta_2, \dots, \eta_{n-r}$  等价。

由于两个向量组均线性无关，所以  $s = n - r$ 。

因为  $\alpha_1, \alpha_2, \dots, \alpha_{n-r}$  可由基础解系  $\eta_1, \eta_2, \dots, \eta_{n-r}$  线性表示，因此  $\alpha_1, \alpha_2, \dots, \alpha_{n-r}$  每个向量都是  $AX = 0$  的解。

对于  $AX = 0$  的任一解  $\eta$ ， $\eta$  可由  $\eta_1, \eta_2, \dots, \eta_{n-r}$  线性表示，而  $\eta_1, \eta_2, \dots, \eta_{n-r}$  可由  $\alpha_1, \alpha_2, \dots, \alpha_{n-r}$  线性表示，则  $\eta$  可由  $\alpha_1, \alpha_2, \dots, \alpha_{n-r}$  线性表示，而  $\alpha_1, \alpha_2, \dots, \alpha_{n-r}$  线性无关，所以由基础解系的定义知， $\alpha_1, \alpha_2, \dots, \alpha_{n-r}$  也是基础解系。